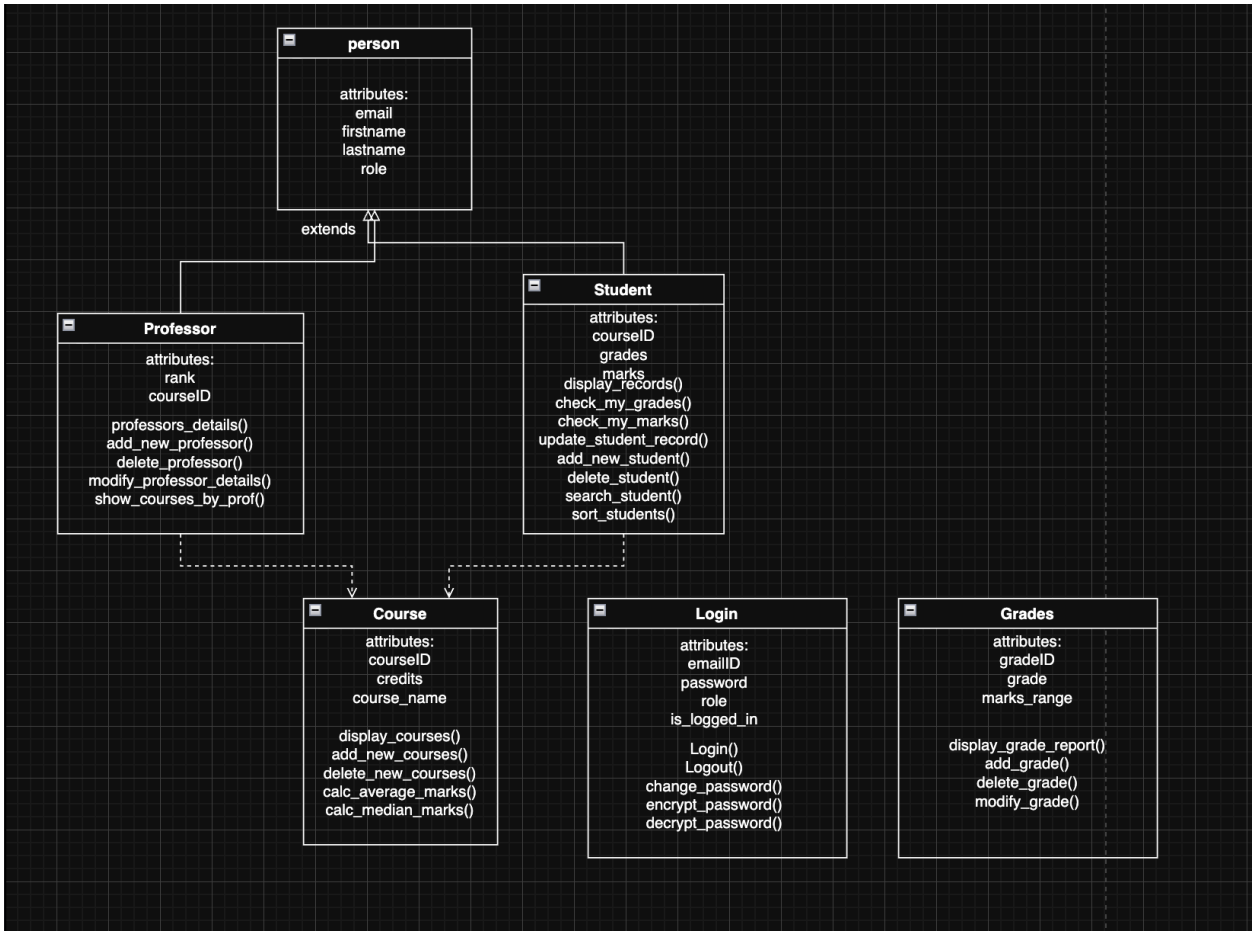
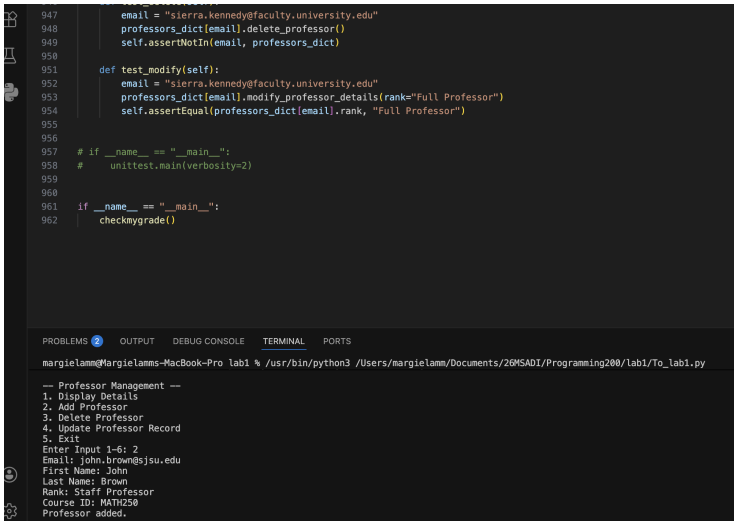
Ethan To DATA 200 Lab 1

Relationship Diagram



Add Professor



```
        email = "sierra.kennedy@faculty.university.edu"
        professors_dict[email].delete_professor()
        self.assertNotIn(email, professors_dict)

    def test_modify(self):
        email = "sierra.kennedy@faculty.university.edu"
        professors_dict[email].modify_professor_details(rank="Full Professor")
        self.assertEqual(professors_dict[email].rank, "Full Professor")


# if __name__ == "__main__":
#     unittest.main(verbosity=2)


if __name__ == "__main__":
    checkmygrade()
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py

-- Professor Management --
1. Display Details
2. Add Professor
3. Delete Professor
4. Update Professor Record
5. Exit
Enter Input 1-6: 2
Email: john.brown@sjsu.edu
First Name: John
Last Name: Brown
Rank: Staff Professor
Course ID: MATH250
Professor added.
```

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View Professor by Email

```
if __name__ == "__main__":
    checkmygrade()
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py
Professor added.

-- Professor Management --
1. Display Details
2. Add Professor
3. Delete Professor
4. Update Professor Record
5. Exit
Enter Input 1-6: 1
Email: john.brown@sjsu.edu
Email:john.brown@sjsu.edu | John Brown | Rank:Staff Professor | CourseID: MATH250

-- Professor Management --
1. Display Details
2. Add Professor
```

Delete Professor by Email

```
-- Professor Management --
1. Display Details
2. Add Professor
3. Delete Professor
4. Update Professr Record
5. Exit
Enter Input 1-5: 3
Enter Professor Email to delete: john.brown@sjsu.edu
Professor john.brown@sjsu.edu | John Brown deleted.
```

Update Professor

```
-- Professor Management --
1. Display Details
2. Add Professor
3. Delete Professor
4. Update Professr Record
5. Exit
Enter Input 1-5: 4
Enter Professor Email to update: claire.scott@sjsu.edu
Leave blank to keep current value.
New First Name (Claire):
New Last Name (Scott):
New Rank (Adjunct Professor): Senior Professor
New Course ID (MATH250):
Details updated for: claire.scott@sjsu.edu Claire Scott

-- Professor Management --
1. Display Details
2. Add Professor
3. Delete Professor
4. Update Professr Record
5. Exit
Enter Input 1-5: 1
Email: claire.scott@sjsu.edu
Email:claire.scott@sjsu.edu | Claire Scott | Rank:Senior Professor | CourseID: MATH250
```

Display Courses

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```
        email = "sierra.kennedy@sjsu.edu"
        professors_dict[email].delete_professor()
        self.assertNotIn(email, professors_dict)

    def test_modify(self):
        email = "sierra.kennedy@sjsu.edu"
        professors_dict[email].modify_professor_details(rank="Test Professor")
        self.assertEqual(professors_dict[email].rank, "Test Professor")


# if __name__ == "__main__":
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py

-- Course Management --
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 1

--- All Courses ---
ID: CS101 | Name: Intro to Computer Science | Credits: 3
ID: CS201 | Name: Data Structures | Credits: 3
ID: CS301 | Name: Algorithms | Credits: 4
ID: CS350 | Name: Operating Systems | Credits: 3
ID: CS410 | Name: Database Systems | Credits: 3
ID: CS450 | Name: Software Engineering | Credits: 4
ID: CS480 | Name: Computer Networks | Credits: 3
ID: MATH101 | Name: Calculus I | Credits: 4
ID: MATH201 | Name: Calculus II | Credits: 4
ID: MATH250 | Name: Linear Algebra | Credits: 3
ID: MATH310 | Name: Probability and Statistics | Credits: 3
ID: BUS101 | Name: Intro to Business | Credits: 3
ID: BUS210 | Name: Accounting Fundamentals | Credits: 3
ID: BUS320 | Name: Business Communication | Credits: 3
```

Add Course

```
        self.assertEqual(professors_dict[email].rank, "Test Professor")


# if __name__ == "__main__":
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py
ID: SCI205 | Name: Chemistry I | Credits: 4
ID: PSY101 | Name: Intro to Psychology | Credits: 3

-- Course Management --
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 2
Course ID: DATA200
Course Name: Programming for Analytics
Credits: 4
Course 'Programming for Analytics' added.
```

Update Course

```
Enter Input 1-7: 4
Enter Course ID to update: DATA200
Leave blank to keep current value.
New Course Name (Programming for Analytics):
New Credits (4): 3
Course DATA200 updated.

-- Course Management --
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 1

--- All Courses ---
ID: CS101 | Name: Intro to Computer Science | Credits: 3
ID: CS201 | Name: Data Structures | Credits: 3
ID: CS301 | Name: Algorithms | Credits: 4
ID: CS350 | Name: Operating Systems | Credits: 3
ID: CS410 | Name: Database Systems | Credits: 3
ID: CS450 | Name: Software Engineering | Credits: 4
ID: CS480 | Name: Computer Networks | Credits: 3
ID: MATH101 | Name: Calculus I | Credits: 4
ID: MATH201 | Name: Calculus II | Credits: 4
ID: MATH250 | Name: Linear Algebra | Credits: 3
ID: MATH310 | Name: Probability and Statistics | Credits: 3
ID: BUS101 | Name: Intro to Business | Credits: 3
ID: BUS210 | Name: Accounting Fundamentals | Credits: 3
ID: BUS320 | Name: Business Communication | Credits: 3
ID: BUS401 | Name: Strategic Management | Credits: 4
ID: ENG101 | Name: English Composition | Credits: 3
ID: ENG210 | Name: Technical Writing | Credits: 3
ID: SCI101 | Name: General Biology | Credits: 4
ID: SCI205 | Name: Chemistry I | Credits: 4
ID: PSY101 | Name: Intro to Psychology | Credits: 3
ID: DATA200 | Name: Programming for Analytics | Credits: 3
```

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Average Marks by Course

```
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 1

--- All Courses ---
ID: CS101 | Name: Intro to Computer Science | Credits: 3
ID: CS201 | Name: Data Structures | Credits: 3
ID: CS301 | Name: Algorithms | Credits: 4
ID: CS350 | Name: Operating Systems | Credits: 3
ID: CS410 | Name: Database Systems | Credits: 3
ID: CS450 | Name: Software Engineering | Credits: 4
ID: CS480 | Name: Computer Networks | Credits: 3
ID: MATH101 | Name: Calculus I | Credits: 4
ID: MATH201 | Name: Calculus II | Credits: 4
ID: MATH250 | Name: Linear Algebra | Credits: 3
ID: MATH310 | Name: Probability and Statistics | Credits: 3
ID: BUS101 | Name: Intro to Business | Credits: 3
ID: BUS210 | Name: Accounting Fundamentals | Credits: 3
ID: BUS320 | Name: Business Communication | Credits: 3
ID: BUS401 | Name: Strategic Management | Credits: 4
ID: ENG101 | Name: English Composition | Credits: 3
ID: ENG210 | Name: Technical Writing | Credits: 3
ID: SCI101 | Name: General Biology | Credits: 4
ID: SCI205 | Name: Chemistry I | Credits: 4
ID: PSY101 | Name: Intro to Psychology | Credits: 3
ID: DATA200 | Name: Programming for Analytics | Credits: 3

-- Course Management --
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 5
Enter Course ID: CS101
Course: Intro to Computer Science | Students: 53 | Average: 72.00
```

Median Marks by Course

```
-- Course Management --
1. Display Courses
2. Add Course
3. Delete Course
4. Update Course
5. Marks Average by Course
6. Median Marks by Course
7. Exit
Enter Input 1-7: 6
Enter Course ID: CS101
Course: Intro to Computer Science | Students: 53 | Median Marks: 71.00
```

View Students with Sort

```
5. Update Student Record
6. Exit
Enter Input 1-6: 1
Sort by: (1) Last Name  (2) Marks  (3) Email
Enter 1, 2 or 3: 2
Sort Ascending (A) or Descending (D)? D

--- Sorted Students ---
ID: chanel.cooper@sjsu.edu | Chanel Cooper | CourseID:BUS210 | Grade: A | Marks:100
ID: reina.burns@sjsu.edu | Reina Burns | CourseID:CS480 | Grade: A | Marks:100
ID: lauren.williams@sjsu.edu | Lauren Williams | CourseID:CS450 | Grade: A | Marks:100
ID: leon.hamilton@sjsu.edu | Leon Hamilton | CourseID:MATH250 | Grade: A | Marks:100
ID: rebecca.flores@sjsu.edu | Rebecca Flores | CourseID:MATH201 | Grade: A | Marks:100
ID: jake.brown@sjsu.edu | Jake Brown | CourseID:BUS210 | Grade: A | Marks:100
ID: sara.palmer@sjsu.edu | Sara Palmer | CourseID:CS480 | Grade: A | Marks:100
ID: miguel.mcdonald@sjsu.edu | Miguel McDonald | CourseID:CS480 | Grade: A | Marks:100
ID: joseph.webb@sjsu.edu | Joseph Webb | CourseID:MATH310 | Grade: A | Marks:100
ID: ana.adams@sjsu.edu | Ana Adams | CourseID:CS350 | Grade: A | Marks:100
ID: jason.bishop@sjsu.edu | Jason Bishop | CourseID:ENG210 | Grade: A | Marks:100
ID: nicole.scott@sjsu.edu | Nicole Scott | CourseID:BUS320 | Grade: A | Marks:100
ID: rachel.rodriguez@sjsu.edu | Rachel Rodriguez | CourseID:CS201 | Grade: A | Marks:100
ID: hunter.porter@sjsu.edu | Hunter Porter | CourseID:CS480 | Grade: A | Marks:100
```

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Search Students

```
-- Student Management --
1. Display Students
2. Search Student
3. Add Student
4. Delete Student
5. Update Student Record
6. Exit
Enter Input 1-6: 2
Enter Student Email to search: jams.evans@sjsu.edu
Student not found.

-- Student Management --
1. Display Students
2. Search Student
3. Add Student
4. Delete Student
5. Update Student Record
6. Exit
Enter Input 1-6: 2
Enter Student Email to search: james.evans@sjsu.edu
ID: james.evans@sjsu.edu | James Evans | CourseID:CS101 | Grade: D | Marks:45
```

Courses by Professor

```
5. Show All Courses by Professor
6. Change My Password
7. Logout
Enter Input 1-8: 5

--- Courses by Professor ---

Professor: Claire Scott | Rank: Senior Professor
ID: MATH250 | Linear Algebra | Credits: 3

Professor: Jasmine Johnson | Rank: Visiting Professor
ID: SCI101 | General Biology | Credits: 4

Professor: Amy Robinson | Rank: Assistant Professor
ID: MATH310 | Probability and Statistics | Credits: 3

Professor: Gary Young | Rank: Visiting Professor
ID: ENG101 | English Composition | Credits: 3

Professor: Ivan Vargas | Rank: Adjunct Professor
```

Change Password

```
-- Professor Menu --
1. Professor Management
2. Course Management
3. Student Management
4. Grade Management
5. Show All Courses by Professor
6. Change My Password
7. Logout
Enter Input 1-8: 6
Enter current password: claires04
Enter new password: claires05
Password updated successfully.
```

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Logout

```
-- Professor Menu --
1. Professor Management
2. Course Management
3. Student Management
4. Grade Management
5. Show All Courses by Professor
6. Change My Password
7. Logout
Enter Input 1-8: 7
claire.scott@sjsu.edu has been logged out.

All data saved to CSV files.

-- Welcome to the Grade Checker --
1. Login
2. Exit
Enter value 1 or 2: 2
Exiting...
margielamm@Margielamms-MacBook-Pro lab1 %
```

Encrypted Password (and login csv)

User_id	Password	Role
claire.scott@sjsu.edu	ktiqzm{8=	professor

Student CSV

Email	First_name	Last_name	Course_id	Grades	Marks
sofia.patel@	Sofia	Patel	BUS210	C	59
andrea.kenn	Andrea	Kennedy	MATH201	B	77
carmen.singh	Carmen	Singh	CS350	C	55
jake.bryant@	Jake	Bryant	BUS320	A	89
tiffany.richar	Tiffany	Richardson	BUS210	A	94
jamie.pierce	Jamie	Pierce	CS350	A	91
aiden.spence	Aiden	Spencer	MATH250	A	87
charlotte.tur	Charlotte	Turner	BUS320	B	71
dillon.mcdon	Dillon	McDonald	CS201	B	75
kaitlyn.boyd@	Kaitlyn	Boyd	CS201	B	80
miranda.ahm	Miranda	Ahmed	MATH101	B	78
erica.gardne	Erica	Gardner	ENG210	C	55

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Course CSV

Course_id	Credits	Course_name
CS101	3	Intro to Computer Science
CS201	3	Data Structures
CS301	4	Algorithms
CS350	3	Operating Systems
CS410	3	Database Systems
CS450	4	Software Engineering
CS480	3	Computer Networks
MATH101	4	Calculus I

Professor CSV

Email	First_name	Last_name	Rank	CourseID
claire.scott@	Claire	Scott	Senior Profes	MATH250
jasmine.john	Jasmine	Johnson	Visiting Profe	SCI101
amy.robinsor	Amy	Robinson	Assistant Pro	MATH310
gary.young@	Gary	Young	Visiting Profe	ENG101
ivan.vargas@	Ivan	Vargas	Adjunct Profe	CS480
riley.reed@s	Riley	Reed	Full Professo	SCI101
serena.jacks	Serena	Jackson	Adjunct Profe	PSY101
kaitlyn.rome	Kaitlyn	Romero	Lecturer	CS480
paul.crawfor	Paul	Crawford	Full Professo	CS301
christopher.s	Christopher	Stephens	Associate Pro	BUS210
joseph.webb	Joseph	Webb	Lecturer	MATH250
ricardo.chen	Ricardo	Chen	Lecturer	MATH310
brooke.garci	Brooke	Garcia	Senior Lectu	CS301
fatima.barne	Fatima	Barnes	Assistant Pro	CS101

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Unit Testing

```python
if __name__ == "__main__":
    unittest.main(verbosity=2)


# if __name__ == "__main__":
#     checkmygrade()
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter value 1 or 2: 2
Exiting...
margielamm@Margielamms-MacBook-Pro lab1 % /usr/bin/python3 /Users/margielamm/Documents/26MSADI/Programming200/lab1/To_lab1.py
test_add (__main__.TestCourse) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_delete (__main__.TestCourse) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_modify (__main__.TestCourse) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_add (__main__.TestProfessor) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_delete (__main__.TestProfessor) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
Professor sierra.kennedy@sjsu.edu | Sierra Kennedy deleted.
ok
```

```
test_modify (__main__.TestProfessor) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
Details updated for: sierra.kennedy@sjsu.edu Sierra Kennedy
ok
test_add (__main__.TestStudent) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_delete (__main__.TestStudent) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
ok
test_modify (__main__.TestStudent) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.
Updated Sofia Patel ID: sofia.patel@sjsu.edu
ok
```

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```
test_search (__main__.TestStudent) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.

Search time across 4 rows: 0.00s
ok
test_sort_students (__main__.TestStudent) ... Courses loaded.
Professors loaded.
Students loaded.
Logins loaded.

lastname ascending:  0.052s
lastname descending: 0.053s
marks ascending:     0.090s
marks descending:    0.091s
email ascending:     0.050s
email descending:    0.050s
ok

----------------------------------------------------------------------
Ran 11 tests in 0.710s

OK
```